# Aaron Espinoza

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## PUBLICATIONS

Enhance Multi-modal and Multi-center Whole Heart Segmentation Using Data Augmentation and Model Calibration. *CARE Workshop, MICCAI 2024.* doi.org/10.1007/978-3-031-87009-5_13

## RESEARCH EXPERIENCE

**Undergraduate Deep Learning Researcher** **Aug. 2024 – Present**
*Texas State University* *San Marcos, TX*

- Research GAN-based adversarial perturbation generation against facial recognition systems, using hybrid U-Net generators to produce perturbations that transfer across models while remaining visually imperceptible.
- Achieved a 98.7% average impersonation success rate across standard benchmarks (LFW, CASIA-WebFace).
- Designed targeted and untargeted attack formulations, balancing match-accuracy degradation against perceptual fidelity, and built the full PyTorch training and evaluation pipeline: preprocessing, adversarial training loop, and attack-success metrics.

**AI Research Intern** **May 2024 – Aug. 2024**
*Guangdong Provincial People's Hospital* *Guangzhou, China*

- Placed 1st globally in the MICCAI 2024 WHS++ whole heart segmentation challenge, competing against teams from research labs and industry.
- Built ensembled PyTorch CNN segmentation models for cardiac CT and MRI, reaching a 94% Dice score.
- Addressed generalizing across imaging modalities and hospital sites with targeted data augmentation and post-hoc model calibration.
- Co-authored the resulting workshop publication, working in a research team against a fixed challenge deadline.

## ENGINEERING EXPERIENCE

**Google Summer of Code Contributor** **May 2025 – Sept. 2025**
*The FreeBSD Project* *Remote*

- Delivered a modular Rust framework for FreeBSD kernel device drivers with safe wrappers over C kernel objects, now used by other contributors.
- Wrote 11 automated ATF/Kyua tests covering stress, concurrency, error handling, and memory leaks, wired into CI — the same reproducibility discipline applied to systems code.
- Continue as an open-source contributor to the FreeBSD src tree (Sept. 2025 – Present). Writeup: gist.github.com/Acesp25 | Project video: youtu.be/y82-t1tDLWg

## SELECTED PROJECTS

**RustKLD** github.com/Acesp25/rustkld
Framework and CI test harness for FreeBSD kernel drivers written in Rust, with contributor documentation.
*Rust, C, FreeBSD, ATF/Kyua.*

**Self-Hosted Compute and Storage** Self-hosted
One Ubuntu and two FreeBSD servers running QEMU VMs, bhyve VMs, and jails for experiment isolation, plus an NFS store reachable through WireGuard tunnels. Used to run training jobs and keep datasets versioned off a laptop.
*QEMU, bhyve, jails, NFS, WireGuard, pf.*

## EDUCATION

**Texas State University** **Expected Dec. 2026**
*B.S. Honors Computer Science; Minors in Applied Mathematics and Chinese* *San Marcos, TX*

- GPA 3.98/4.00 overall, 4.00/4.00 College of Science & Engineering GPA. Dean's List/President's List, all semesters.
- Extracurriculars: Vice President, TXST Fencing Club (40+ members), TXST Esports Club CS2 Captain.
- Coursework: Linear Algebra, Probability & Statistics, Data Structures & Algorithms.

## SKILLS

**Machine Learning:** PyTorch, TensorFlow, CNNs, U-Nets, GANs, adversarial attacks, image segmentation, ensembling, calibration, data augmentation
**Languages:** Python, C, Rust, Java, C++, Bash
**Math:** Linear algebra, probability and statistics, numerical methods (Applied Mathematics minor)
**Tooling:** Git, Linux, FreeBSD, Anaconda, Jupyter, VS Code, Vim, LaTeX, bhyve, jails, NFS